

NAME

landlock_add_rule – add a new Landlock rule to a ruleset

LIBRARY

Standard C library (*libc*, *-lc*)

SYNOPSIS

```
#include <linux/landlock.h> /* Definition of LANDLOCK_* constants */
#include <sys/syscall.h> /* Definition of SYS_* constants */

int syscall(SYS_landlock_add_rule, int ruleset_fd,
            enum landlock_rule_type rule_type,
            const void *rule_attr, uint32_t flags);
```

DESCRIPTION

A Landlock rule describes an action on an object. An object is currently a file hierarchy, and the related filesystem actions are defined with a set of access rights. This **landlock_add_rule()** system call enables adding a new Landlock rule to an existing ruleset created with **landlock_create_ruleset(2)**. See **landlock(7)** for a global overview.

ruleset_fd is a Landlock ruleset file descriptor obtained with **landlock_create_ruleset(2)**.

rule_type identifies the structure type pointed to by *rule_attr*. Currently, Linux supports the following *rule_type* value:

LANDLOCK_RULE_PATH_BENEATH

This defines the object type as a file hierarchy. In this case, *rule_attr* points to the following structure:

```
struct landlock_path_beneath_attr {
    __u64 allowed_access;
    __s32 parent_fd;
} __attribute__((packed));
```

allowed_access contains a bitmask of allowed filesystem actions for this file hierarchy (see **Filesystem actions in landlock(7)**).

parent_fd is an opened file descriptor, preferably with the *O_PATH* flag, which identifies the parent directory of the file hierarchy or just a file.

flags must be 0.

RETURN VALUE

On success, **landlock_add_rule()** returns 0.

ERRORS

landlock_add_rule() can fail for the following reasons:

EOPNOTSUPP

Landlock is supported by the kernel but disabled at boot time.

EINVAL

flags is not 0, or the rule accesses are inconsistent (i.e., *rule_attr->allowed_access* is not a subset of the ruleset handled accesses).

ENOMSG

Empty accesses (i.e., *rule_attr->allowed_access* is 0).

EBADF

ruleset_fd is not a file descriptor for the current thread, or a member of *rule_attr* is not a file descriptor as expected.

EBADFD

ruleset_fd is not a ruleset file descriptor, or a member of *rule_attr* is not the expected file descriptor type.

EPERM

ruleset_fd has no write access to the underlying ruleset.

EFAULT

rule_attr was not a valid address.

STANDARDS

Linux.

HISTORY

Linux 5.13.

EXAMPLES

See **landlock(7)**.

SEE ALSO

landlock_create_ruleset(2), **landlock_restrict_self(2)**, **landlock(7)**